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Research Article

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Temperature-driven cotton verticillium wilt: a beta model for risk assessment from laboratory insights to climate scenarios

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Abstract

BACKGROUND: Verticillium wilt is a critical disease affecting cotton in the Xinjiang province, a region producing 90% cotton in China. Defining the specific temperature thresholds for disease prevalence is essential but has remained unclear.

RESULTS: This study aimed to establish a model to quantify the relationship between temperature and cotton verticillium wilt disease risk. Through a controlled temperature experiment, we identified a nonlinear temperature relationship, with an optimal temperature of 26.5 °C. Then a beta model, parameterized from these findings, was validated against historical regional disease data, confirming its ability to accurately reflect interannual variations in disease occurrence and its direct applicability from laboratory to regional scales. We then utilized the model to project future disease risks under two Shared Socioeconomic Pathways (SSP) climate scenarios. The projections estimate a 4.8% to 10.1% increase in disease risks in Xinjiang by the 2080s under SSP1-2.6 and SSP5-8.5 scenarios, respectively.

CONCLUSION: This research offers a valuable predictive tool for cotton verticillium wilt risks, informing strategic decisions for cotton production in the face of climate change. The successful application of a laboratory-derived model at a regional scale marks a significant advancement in plant disease risk assessment, underscoring temperature as a dominate factor in cotton disease dynamics.
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Keywords: verticillium wilt; laboratory; Beta; model validation; climate change; China

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1 INTRODUCTION

Verticillium wilt is a pervasive soilborne fungal disease that poses significant challenges to agricultural productivity due to the long-term survival of its microsclerotia in the soil.[1] This disease threatens the yield of numerous crops, leading to substantial economic losses.[2] Cotton, one of the most crucial natural textile crops, is under serious threat from Verticillium wilt in China.[3–5] Recent research has identified temperature as a key factor that may trigger disease outbreaks.[6,7]

The Xinjiang Uyghur Autonomous Region stands as the preeminent cotton-producing area in China, accounting for 85% of the nation's cotton-growing areas and 90% of its cotton production.[8] However, Xinjiang is undergoing a rapid process of climatic warming, with temperatures increasing at a rate of 0.28 °C per decade since 1961.[9–11] Given these changes, understanding the potential impacts of warming on cotton production in Xinjiang is critical for ensuring future cotton yields and agricultural security.

While previous studies have extensively explored the abiotic stresses induced by climatic warming, such as heat stress[12–15] or drought,[16–19] the effects of temperature on plant diseases have received less attention, yet they are a crucial component in climatic risk assessments.[7,20,21] This is because the optimal temperature ranges for many plant diseases are often lower than those causing crop heat injury,[6,22] and farmers may face conditions favoring disease outbreaks earlier than those leading to heat stress. Due to limited disease data, researchers have primarily relied on statistical links between rising temperatures and plant disease spread[7,21,23] or have calibrated models based on field-level observations.[20] However, temperature thresholds and simulation model derived from experiments remains lacking. There is a need for models that can predict disease risk at a regional scale.

Therefore, this study aims to investigate and model the effects of temperature on cotton verticillium wilt disease risks in Xinjiang from laboratory level to regional scale. We seek to (i) identify key cardinal temperature thresholds for cotton verticillium wilt disease in a temperature-control experiment; (ii) validate the regional applicability of the laboratory-derived model; (ii) project future disease risks under various socioeconomic scenarios. Through this research, we aim to provide a foundation for developing more effective and proactive strategies to manage and mitigate the impacts of cotton verticillium wilt disease, thereby safeguarding Xinjiang's cotton production in the face of climate change.

2 MATERIALS AND METHODS

2.1 Fungal isolates and culture conditions

A virulent defoliating verticillium dahliae isolate from cotton in Xinjiang, China, was used in this study. This isolate and its transformants were stored at −80 °C in the form of microconidial suspension in 20% glycerol. Cultures were reactivated on Potato Dextrose Agar (PDA) medium (Becton, Dickinson and Company). Microsclerotial formation assays were performed on infertile nutrient medium, that overlayed a cellulose membrane on basal agar medium supplemented with 20 mM sodium nitrate to create a condition simulated near-synchronous development of melanized microsclerotia. Conidia production for infection assays and germination tests were cultured in liquid Czapek-Dox medium (30 g/L Sucrose, 3 g/L $NaNO_3$, 0.5 g/L $MgSO_4$-$7H_2O$, 0.5 g/L KCl, 100 mg/L $FeSO_4$-$7H_2O$, 1 g/L K_2HPO_4, pH 7.2).

2.2 Temperature control experiment

In order to investigate the effects of temperature on the progression of Verticillium dahliae in cotton plants, we conducted a controlled experiment within climate chambers. The cotton plants were inoculated using a 'laboratory unimpaired root dip-inoculation method'.[24] The cotton variety selected for this experiment is Xinluzao-33, a cultivar that has been introduced and subsequently gained widespread adoption in Xinjiang since 2010. Fungal cultures, which had been incubated for 5 days in liquid Czapek-Dox medium, were filtered through several layers of cheesecloth to separate the conidia from the mycelia. The concentration of conidia was then adjusted to approximately 1×10^7 conidia per mL. 12. A total of 12 seedlings per pot were planted in MS liquid medium (Murashige and Skoog medium) and were grown in an environmentally controlled growth room. The conditions in the growth room were maintained at 26 °C with 60–70% relative humidity, under a 16-h light/8-h dark photoperiod, for an initial 2-week period. After this acclimatization, the seedlings were inoculated by immersing their roots in the conidial suspension for a duration of 50 min. Subsequently, the inoculated seedlings were transplanted back into 1/10 MS liquid medium and placed in climate chambers for further growth under five different temperature treatments, i.e., 19 °C, 22 °C, 25 °C, 28 °C, and 31 °C. This treatment setting allows us to capture the temperature effect on the disease while excluding the potentially confounding effects, like heat.[15] Each temperature treatment was replicated three times. The disease progress in the cotton plants was monitored until it reached a plateau. The cotton disease index was quantified following the standard technical specification for evaluating resistance of cotton to diseases and insect pests, as outlined in Part 5 of the GB/T22101 standard.[25] The difference between temperature treatments was determined by the Tukey HSD Test.

2.3 Modelling and validation at regional scale

Based on the above experiment, we modelled the effect of temperature on disease index using a beta model (Eqns 1 and 2):

$$DI = DI_{max} \cdot r(T) \tag{1}$$

$$r(T) = \left[\left(\frac{T - TB}{TO - TB} \right) \left(\frac{TM - T}{TM - TO} \right)^{\frac{TM - TO}{TO - TB}} \right]^{ALPHA} \cdot 100\% \tag{2}$$

where DI is the cotton disease index (%), DI_{max} is the maximum disease index (%); $r(T)$ is temperature-dependent disease risk for cotton verticillium wilt disease, varying from 0 to 100%; T is temperature in treatment (°C); relative values of $r(T)$ can be calculated by a beta function with the base (TB), optimum (TO) and maximum (TM) cardinal temperature (°C); ALPHA is an empirical parameter that defines the sensitivity of the disease with temperature (−). With our experimental data, all parameters can be determined by a nonlinear least-squares algorithm *nlsLM* from *minpack.lm* software package.[26] The beta model can be viewed as the temperature effect on cotton verticillium wilt disease under a laboratory environment.

Next, we validated the relative disease risk, $r(T)$, obtained under a laboratory environment by the regional cotton disease statistics in Xinjiang (Fig. 1). We collected the occurrence area of cotton verticillium wilt disease in Xinjiang from the National Plant Protection Statistics in China, and the cotton planting areas in Xinjiang from the National Bureau of Statistics over the period of 2011–2021. By

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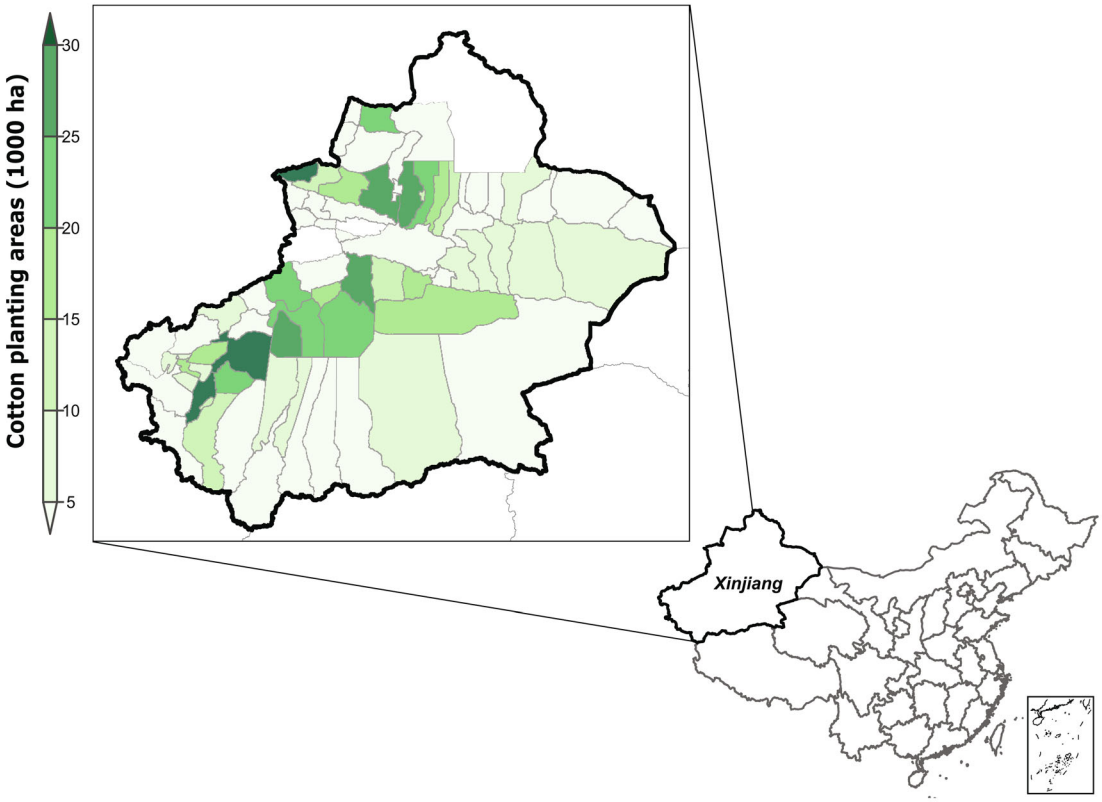


Figure 1. Study region of Xinjiang, China and the cotton planting area in each county.

integrating these two datasets, we calculated the annual ratio of the occurrence area of cotton verticillium wilt disease in Xinjiang. This ratio, represented by Eqn. 3, provides a normalized measure of the disease's spread, allowing for a comparison between the laboratory-based disease risk model and the actual disease incidence patterns observed in the field.

$$RO_y = \frac{Occurence\ Area_y}{Planting\ Area_y} \times 100\% \quad (3)$$

where RO is the ratio of occurrence area of cotton verticillium wilt disease; *Occurrence Area* is the area that cotton verticillium wilt disease occurred; *Planting Area* is the cotton growing area in the year of *y* in Xinjiang.

In order to apply the beta model at a regional scale within Xinjiang province of China, we first downloaded a high-resolution climate reanalysis dataset that provides gridded climate data for China,[27] sourced from the Climate Change Research Center at the Chinese Academy of Sciences (https://ccrc.iap.ac.cn/resource/detail?id=228). The dataset encompasses daily records of minimum and maximum temperatures. We then fitted a cosine curve (Eqn. 4) to extrapolate the daily temperature to hourly temperature over the cotton growing season in Xinjiang,[28] which were then input into the above beta model (Eqn. 1) to calculate the average r_c over the growing season for each county and each year. To aggregate the disease risk to a regional level and estimate the overall risk for cotton verticillium wilt disease in Xinjiang, we weighed the cotton planting area of each county by its corresponding r_c. This county-level risk was then normalized by dividing it by the total cotton planting area across Xinjiang (Eqn. 5).

$$T_h = \frac{T_{min} + T_{max}}{2} + (T_{max} - T_{min})\frac{\cos(0.2618(h-14))}{2} \quad (4)$$

$$r_{reg} = \frac{\sum r_c \cdot \mathrm{Area}_c}{\sum \mathrm{Area}_c} \quad (5)$$

where T_h is the hourly temperature; T_{min} and T_{max} is the daily minimum and maximum temperatures; h is the hours from 1 to 24; r_{reg} is the regional disease risks of Xinjiang; r_c is the disease risk in certain county; Area_c is the cotton planting area in that county.

The ratio of occurrence area of cotton disease, derived from official statistics, serves as an empirical measure of the actual disease severity, reflecting the combined effects of various factors on the disease's prevalence. In contrast, the regional disease risk represents a theoretical index that specifically considers the influence of temperature on the disease's development. While we did not anticipate that the regional disease risk could accurately reproduce the ratio of occurrence area, the synchronicity between these two indicators could offer valuable insights. If the regional disease risk aligns with the fluctuations in the ratio of occurrence area, it could serve as a useful proxy indicator for the occurrence of cotton verticillium wilt disease in Xinjiang. This alignment would validate the regional disease risk as a predictive tool at a regional scale.

To test this hypothesis, we analyzed the ratio of occurrence area of cotton verticillium wilt and the regional disease risk over the

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period from 2011 to 2021. Firstly, we detrended both variables by fitting a cubic smoothing spline, a technique that smooths the data to remove non-climatic factors (such as disease-resistant cotton variety). After that, we compared the interannual deviations of the two variables to assess their synchronicity. We defined a successful prediction as instances where both variables estimate a positive (or negative) interannual deviation in the same direction. Then the predication accuracy (Eq. 6) will be calculated as the proportion of years with right predictions relative to the total number of years.

$$PA = \frac{N_{\text{right prediction}}}{N_{\text{total prediction}}} \quad (6)$$

where PA is prediction accuracy (–); $N_{\text{right prediction}}$ is number of years to make right prediction, i.e., both positive or negative interannual deviation for observations and predictions; $N_{\text{total prediction}}$ is the total years to make prediction.

2.4 Projected disease risk change under future warmer climate

With the above-validated model, we project the climate change impacts on disease risk of cotton verticillium wilt disease in Xinjiang in two Shared Socioeconomic Pathways (SSP), i.e., SSP1-2.6 and SSP5-8.5 scenarios. The two scenarios represent the lowest and the highest global greenhouse gas emission scenarios; SSP1-2.6 represents a sustainable socioeconomic development pathway, and SSP5-8.5 is a high greenhouse gas emissions scenario, corresponding to a future that avoids at all climate change mitigation costs.[29] Five general circulation models (GCMs), i.e., GFDL-ESM4,[30] IPSL-CM6A-LR,[31] MPI-ESM-2-HR,[32] MRI-ESM2-0,[33] and UKESM1–0-LL,[34] were downloaded (https://www.isimip.org/gettingstarted/input-data-bias-adjustment/). The future climate model projections were processed using Eqns 4 and 5 to estimate the potential changes in the risk of cotton Verticillium wilt disease due to future warming in Xinjiang. We projected the disease risks using each individual model and then calculated the average and 95% confidence intervals across all five models.

3 RESULTS

3.1 Nonlinear disease response with temperature

The results of our temperature control experiment indicated that the progression of cotton verticillium wilt disease is influenced by temperature in a nonlinear manner, as shown in Fig. 2. The disease exhibited more rapid development at temperatures ranging from 25 to 28 °C, with the rate of progression noticeably decelerating outside of this temperature range. No statistically significant differences were observed in the disease index between the 25 °C and 28 °C treatments. However, a significant reduction in the disease index was noted under temperatures of 19 °C and 23 °C, with this decrease being statistically significant ($P < 0.05$). Similarly, the disease index at a temperature of 31 °C was lower compared to those at 25 °C and 28 °C with statistical significance ($P < 0.05$).

Based on these experimental data, we quantitatively estimated beta model parameters that describe the relationship between temperature and the response of cotton verticillium wilt disease index (Fig. 2). The model revealed that the optimal cardinal temperature (TO) is 26.5 °C, in which the disease index (DI) is predicted to reach its maximum value (DI_{max}) of 82.01%. The base (TB) and maximum cardinal temperatures (TM), which represent

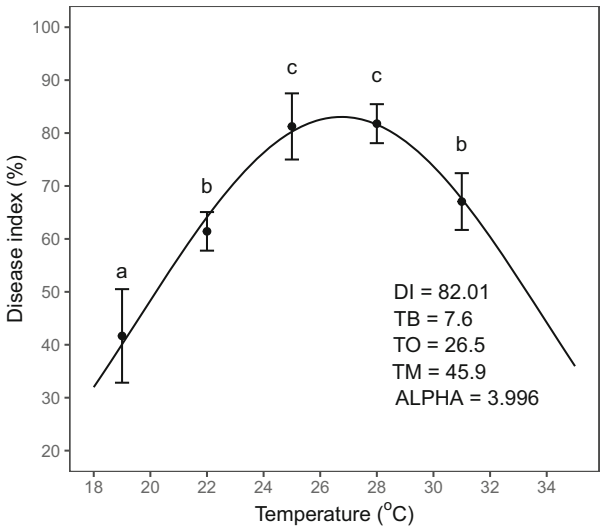


Figure 2. The nonlinear change of cotton disease index with temperature and the fitted beta model based on the experimental data. The data point is the results of temperature-control experiment; the error bars indicate the variance between replicants; letters indicate statistical significance among temperature treatments using two-sided Tukey HSD Test ($P < 0.05$); the solid line is the beta curve estimated based on experiment; the model parameters are listed in the plot.

the lower and upper thresholds for disease progression, are determined to be 7.6 °C and 45.9 °C, respectively. At these extreme temperatures, the disease progression is expected to cease entirely. The ALPHA parameter, which is calculated to be 3.996, plays a crucial role in defining the curvature of the beta curve, thereby reflecting the rate of change in disease risk with respect to temperature fluctuations. This inverse-U shaped response to temperature is consistent with theoretical plant pathogen models.[6]

3.2 Modelling validation at regional scale

Using the beta model, which was parameterized under controlled laboratory conditions, we estimated the regional disease risks for cotton verticillium wilt disease in Xinjiang. These estimated risks were then juxtaposed with the annual occurrence ratio of the diseased area (Fig. 3). Our analysis revealed a strong correlation between the regional disease risk (indicated by the blue line in Fig. 3(a)) and the interannual variability in the occurrence of cotton verticillium wilt disease (indicated by the red line in Fig. 3(a)). For instance, both the regional disease risk and the occurrence ratio exhibited a decline in 2014 compared to the surrounding years. Additionally, the years between 2018 to 2020 marked another relatively lower period in the wilt disease (Fig. 3(a)).

After detrending the time series, the interannual deviations in regional disease risks (indicated by the blue bars in Fig. 3(b)) were found to be largely synchronous with the occurrence area ratio (indicated by the red bars in Fig. 3(b)) in most years. Notably, the year 2017 deviated from this pattern, with the regional disease risk indicating a higher-than-expected risk, while the actual occurrence of cotton verticillium wilt disease was lower than predicted. Over the 11-year period from 2011 to 2021, this resulted in 10 successful predictions and a single failed prediction, yielding a prediction accuracy (PA) of 0.909.

Furthermore, our analysis indicated that for every 1 % increase in the regional disease risk, there is a tendency for the occurrence

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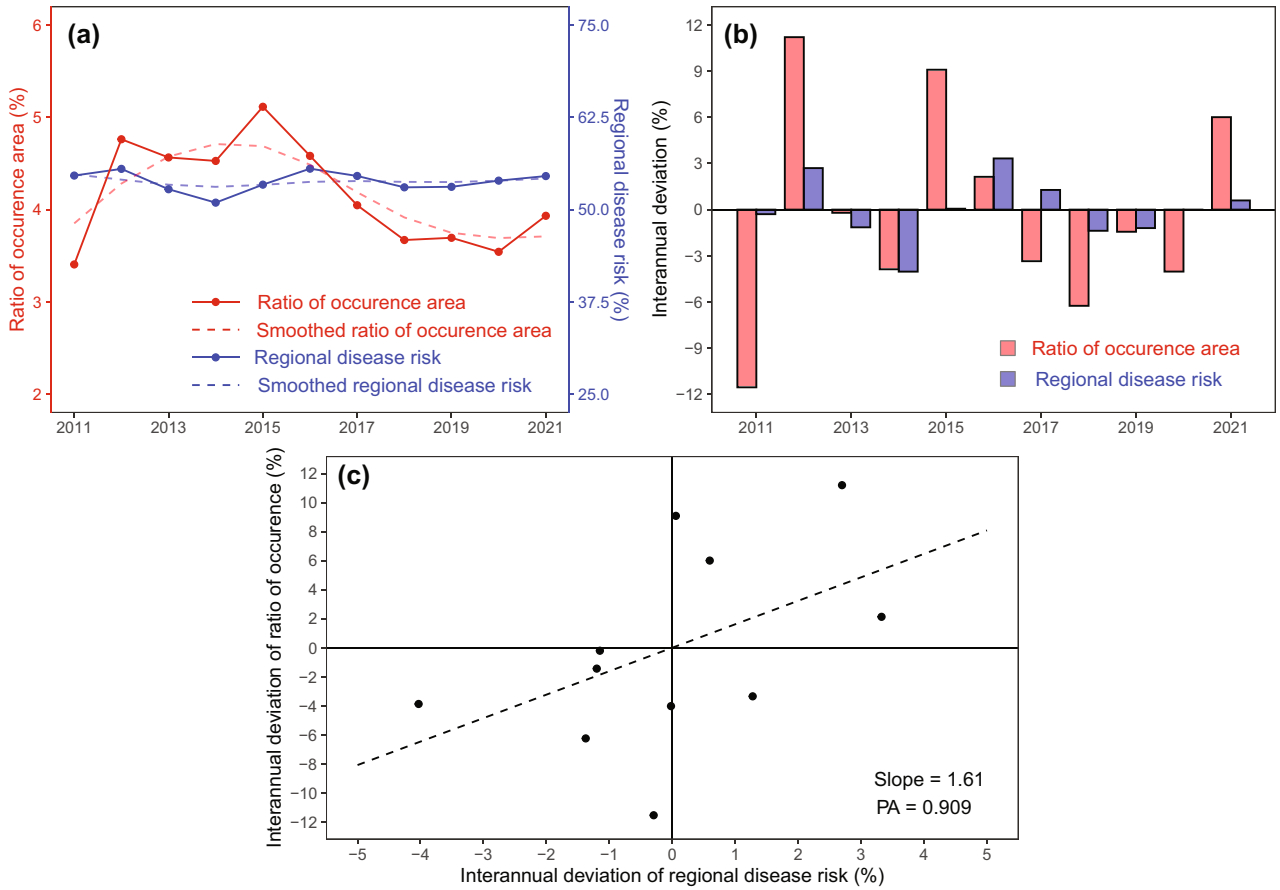


Figure 3. Comparison between the ratio of occurrence area of cotton verticillium wilt disease and regional disease risks estimated based on temperature. (a) Yearly data of the ratio of occurrence area of cotton verticillium wilt disease and regional disease risks; (b) the interannual deviation of the ratio of occurrence area of cotton verticillium wilt disease and regional disease risks after detrending time series; (c) comparison between the two variables, and the dot line is the regression line between the variables.

area ratio of cotton verticillium wilt disease to rise by 1.61%, as indicated by the slope in Fig. 3(c). This finding underscores the potential utility of the beta model in predicting the spatial distribution and prevalence of the disease.

3.3 Potential impact of future warming on cotton verticillium wilt disease risks

Building on the validated model, we projected the potential changes in cotton verticillium wilt disease risks due to future climatic warming, in comparison to the baseline climate, under both the SSP1-2.6 and SSP5-8.5 scenarios, as depicted in Fig. 4. There is an increase of up to 1.2 °C in both minimum and maximum temperatures for the SSP1-2.6 scenario. In contrast, the SSP5-8.5 scenario projects a temperature increase of around 5.8 °C by the 2080s (Fig. S1). The projections indicate that disease risks are expected to increase in Xinjiang under both climate scenarios, with the risks under the SSP5-8.5 scenario being notably higher than those under SSP1-2.6. By the 2080s, the model anticipates an increase in disease risk of approximately 4.8% under the SSP1-2.6 scenario, whereas a more substantial rise of about 10.1% is projected under the SSP5-8.5 scenario for the same time, as shown in Fig. 4(a). Meanwhile, the increasing rates are also gradually reduced for the two scenarios (Fig. 4(a)). Under SSP1-2.6, the majority of counties are expected to face elevated disease risks during the 2080s. However, the spatial distribution of these risks is less distinct (Fig. 4(b)). In contrast, the spatial distribution of disease risks under the SSP5-8.5 scenario is more pronounced, with a discernible pattern in 2080s. The model predicts that the western counties will experience a more rapid increase in disease risks compared to those in central Xinjiang (Fig. 4(c)). A notable exception is a single county located in the southeast of Xinjiang, which also exhibits a significant rise in disease risks.

Figure 5 presents an analysis of county-specific temperatures in the baseline climate, trends of temperature over the cotton growing season, and the estimated trends of disease risk under both SSP1-2.6 and SSP5-8.5 scenarios. Under the SSP1-2.6 scenario, our analysis indicates that counties with an average baseline temperature in the range of 16 to 20 °C exhibit a more rapid increase in disease risks compared to other counties (Fig. 5(a)). Conversely, for counties with baseline temperatures above 24 °C or below 14 °C, the disease risk associated with cotton verticillium wilt appears to be constrained, resulting in a slower rate of increase compared to those with temperatures within the 16 to 20 °C range. In the SSP5-8.5 scenario, which is characterized by more pronounced warming trends than SSP1-2.6, the temperature range associated with the fastest increasing trends of disease risks shifts to a lower band of 13–17 °C (Fig. 5(b)). Additionally, the model suggests that the development of cotton verticillium wilt disease slows down significantly for counties with baseline temperatures exceeding 20 °C.

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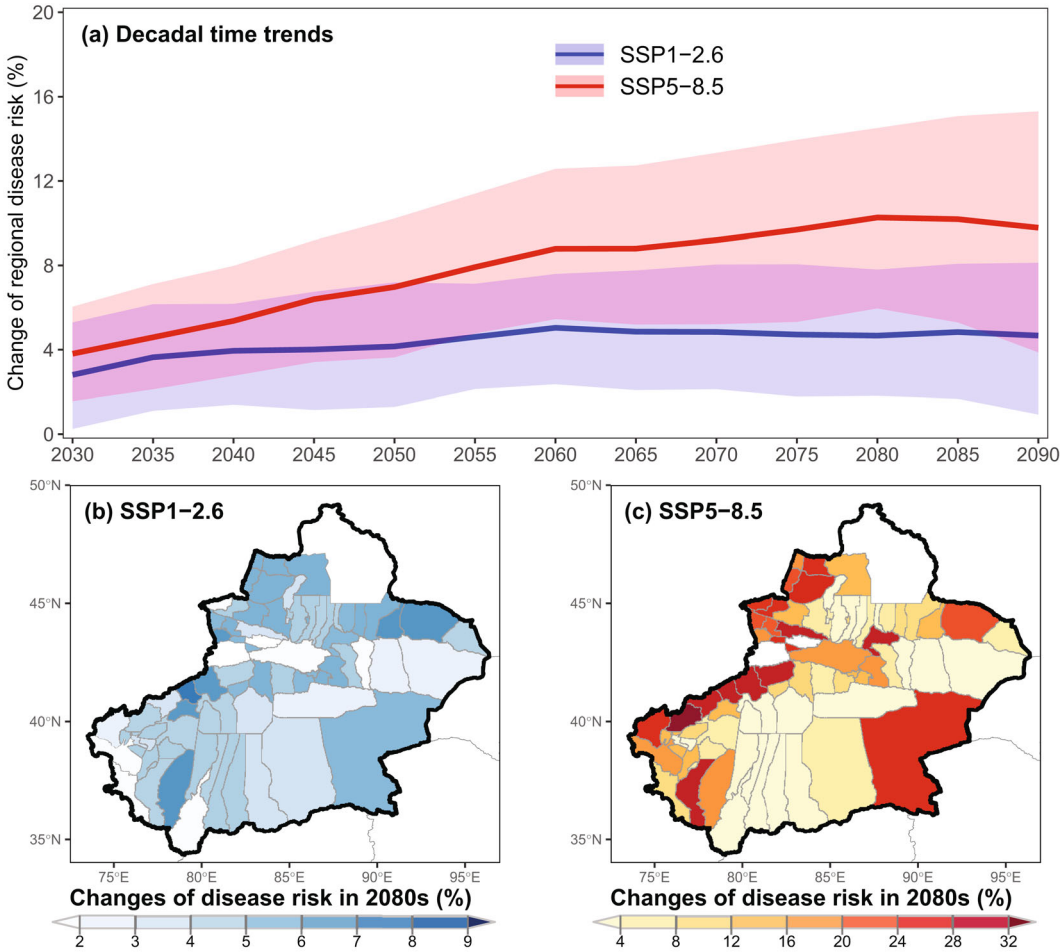


Figure 4. Model projection of potential disease risks changes due to climatic warming relative to the baseline period. Figure 4(a) shows the estimated changes in the area-weighted risks in each future year under the SSP1-2.6 (blue line) and SSP5-8.5 (red line) scenario, and the shading areas are the 95% confidence intervals between GCMs; Fig. 4(b),(c) shows the spatial distribution of risk changes averaged in 2080s under SSP1-2.6 and SSP5-8.5.

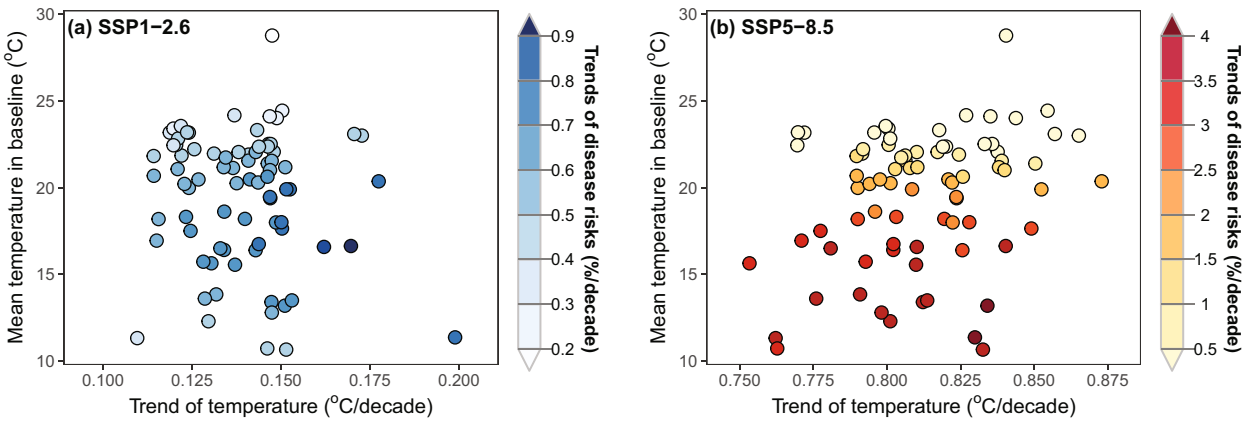


Figure 5. The county-specific trends of disease risks with temperature at baseline climate and temperature trends over cotton growing season under SSP1-2.6 (Fig. 5(a)) and SSP5-8.5 (Fig. 5(b)).

4 DISCUSSION

This study explored the impact of temperature on cotton verticillium wilt disease risks in Xinjiang, China, scaling from laboratory conditions to regional analysis. Our findings align with previous research[10] that identified rising temperatures as a significant factor contributing to the spread of crop diseases in China. However, our study extends this understanding by revealing specific temperature thresholds for cotton verticillium wilt disease, providing a mechanistic insight that goes beyond statistical correlations. Our laboratory-based temperature-control experiment and

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subsequent beta model analysis suggest an optimum cardinal temperature of 26.5 °C for cotton verticillium wilt disease. This closely matches several earlier studies based on either control experiments[6,22] and statistical methods,[21] which identified 20–28 °C as the optimum cardinal temperature for plant disease caused by verticillium dahliae. A significant strength of our study lies in the direct applicability of our laboratory-derived beta model to a regional scale, eliminating the need for further calibration. By further incorporating regional temperature data into our model, we were able to inform the interannual fluctuations and trends in the cotton verticillium wilt disease area in Xinjiang across most years (Fig. 3). This suggests that our simulated disease risk index can be served as a reliable indicator for the interannual changes of actual disease prevalence. It is important to note, however, that our regional validation may be constrained by the relatively short period of available observations for cotton verticillium wilt occurrence. Prior to 2011, only general agricultural pest and disease occurrence data were available in China, lacking specific statistics for individual crops and specific diseases. Additionally, our model is also restricted by our experiment with a single cotton variety, which does not account for genetic variation among cotton plants. While models incorporating a broader range of experiments with multiple varieties and diverse fungal strains are desirable, we emphasize that our current model focuses on interannual scales, with non-climatic factors largely removed through detrending of the time series (Section 2.3).[35]

While our model primarily considers temperature, it is acknowledged that moisture also plays a role in plant disease.[36,37] However, the synchronous fluctuation between simulated temperature-dependent disease risks and actual occurrence area ratios (Fig. 3) suggests that temperature may be the dominant interannual climate driver for this disease in Xinjiang. This is supported by field-level[38] and global-scale[39] studies, which emphasize the primary role of temperature in fungal distribution and activity. Another factor to consider is the management of drip irrigation in Xinjiang's cotton fields, which results in a very uniform spatial distribution of soil moisture.[40] This characteristic could partially explain why temperature seems to be a more significant factor than moisture in driving the interannual variability in disease incidence, a finding that aligns with the results reported by Gui et al.[40]

Using our regionally-validated model, we project an increase in cotton verticillium wilt disease risks due to future warming, with a 4.8% increase under the SSP1-2.6 scenario and a 10.1% increase under SSP5-8.5 by the 2080s (Fig. 4(a)). This indicates that an unsustainable socioeconomic pathway would lead to a warmer climate and a greater spread of cotton diseases. As countries exceed the optimum cardinal temperature, the rate of increase in disease risks is expected to decline. Counties with an average temperature of 13–20 °C during the cotton growing season are identified as most at risk (Fig. 5). While counties with temperatures above 20 °C may experience slower disease spread, we need to caution the heat stresses on cotton growth and production in these countries.[12,14,21] Cotton disease response to temperature may be superposed on the abiotic climatic stresses, reducing yields at lower temperature regimes (26.8 °C) before reaching the critical temperature (32 °C) that causes cotton heat injury.[15,41] Cotton farmers must therefore prepare for and counteract the increasing disease pressure in Xinjiang if the model projections of the study are to be realized.[24,42,43]

Our study establishes key temperature thresholds for Verticillium wilt disease in cotton through controlled laboratory experiments and validates these findings at a regional scale. This research methodology could be applied to identify temperature thresholds for other plant diseases. Future experiments could focus on determining the precise modeling parameters within the beta model for various plant diseases.

5 CONCLUSION

In summary, our research provides a comprehensive examination of the effects of temperature on cotton verticillium wilt disease, with the beta model established from laboratory experiments effectively explaining the nonlinear disease response to temperature. The model also successfully informs the interannual fluctuations in actual cotton verticillium wilt occurrence in Xinjiang. Furthermore, as climatic warming is expected to elevate disease risks, the breeding of disease-tolerant cotton varieties emerges as a critical technological investment for future cotton production. Our study not only estimates the long-term risks of climate change on cotton verticillium wilt disease but also suggests a potential application of the model for seasonal climate forecast data. By feeding in seasonal climate forecast data (e.g., up to 6 months climate forecast), the model can make predictions on cotton verticillium wilt disease before the planting of cotton each year. This could enable predictive disease management for cotton farmers, guiding the selection of disease-resistant varieties and risk mitigation strategies.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in ISIMIP at https://www.isimip.org/gettingstarted/input-data-bias-adjustment/.

CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

SUPPORTING INFORMATION

Supporting information may be found in the online version of this article.

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